2.3 Functional Requirements

Requirement	Description
FR1	Read aircraft data from a captured replay file or a live PingStation TCP stream.
FR2	Decode identity, position and motion fields when those fields are available.
FR3	Filter unusable records, including records with both coordinates equal to zero.
FR4	Expose aircraft, status, logging and trajectory data through JSON API endpoints.
FR5	Display aircraft on an interactive map with callsign and selected-aircraft details.
FR6	Store current aircraft records and meaningful historical trajectory samples locally.
FR7	Support CSV and ICAO-wise Excel-compatible exports for later analysis.

2.4 Non-Functional Requirements

- The aircraft display should refresh without a full page reload.
- The application should run locally without a separate database server.
- Decoder, preprocessing, storage and interface modules should remain independently maintainable.
- Repeated observations should not create unnecessary trajectory samples.
- Predicted records must be visibly distinguished from received records.

2.5 Hardware and Software Requirements

Category	Requirement
Processor	Intel/AMD-compatible processor capable of running Python and a modern browser.
Memory	Minimum 4 GB RAM recommended for the local backend and browser dashboard.
Operating system	Windows 10/11 or another operating system supported by Python and Flask.
Backend	Python, Flask, SQLite, requests and openpyxl.
Frontend	HTML, CSS, JavaScript, Leaflet, Chart.js and OpenStreetMap tiles.
Data	Captured PingStation logs, SQLite database, CSV exports and ICAO workbooks.